

NARASIMHA REDDY

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SUMMARY

Results-driven Healthcare Data Analyst with 4+ years leading transformational analytics initiatives that reduced 30-day hospital readmissions by 17% within 6 months, uncovered \$4.3M+ in fraud annually, and accelerated care coordination effectiveness by 21%. Master architect of HIPAA-compliant AI/ML pipelines, delivering predictive insights and automated ETL solutions that supercharge clinical decision-making across multi-hospital ecosystems. Expert in FHIR/HL7 integration, risk adjustment modeling, and population health analytics, driving scalable, actionable healthcare intelligence to power value-based care and equity. Proven leader and mentor, fostering cross-functional collaboration and building high-performing analytics teams to advance data-driven culture.

SKILLS

Programming & Analytics: SQL (Advanced), Python (Pandas, Scikit-learn, XGBoost, TensorFlow), R, PySpark, NLP (spaCy, NLTK)

Data Engineering & Big Data: Azure Data Factory, Databricks, ETL Automation, API Integration, Hadoop, Kafka, Apache Spark, Azure, AWS HealthLake, GCP

BI & Visualization: Tableau (Certified Specialist), Power BI, Sisense, Dash

Healthcare Systems & Standards: Epic Clarity, Cerner, Athenahealth, FHIR/SMART on FHIR, HL7, ICD-10, CPT,

LOINC, SNOMED CT, OMOP CDM, DICOM

Compliance & Security: HIPAA, HITECH, GxP, HITRUST CSF, NIST, Data Privacy, OAuth2

Modeling & Methodologies: Predictive Modeling, Clustering (DBSCAN), Time Series, Fraud Detection, Risk Adjustment (RAF/HCC)

DevOps & Collaboration: Git, Jira, Confluence, Agile/Scrum, Docker, Kubernetes, CI/CD (Jenkins, Azure DevOps), Terraform

EXPERIENCE

Healthcare Data Analyst | UnitedHealth Group | CT

May 2024 – Present

- Developed and deployed an XGBoost risk prediction model integrating clinical and SDOH data, cutting 30-day hospital readmissions by 17% and boosting care coordination efficiency by 21%, positively impacting 10,000+ patients and driving estimated annual cost savings of \$2.1M.
- Architected scalable, fully automated FHIR/HL7-compliant ETL pipelines using Azure Data Factory and PySpark in a cloud environment, harmonizing Epic and Cerner EMR data across 5 hospital networks—accelerating data availability by 40% and improving ICU operational efficiency by 15%.
- Designed and maintained CI/CD pipelines with Azure DevOps for continuous, error-free deployment of analytics workflows, improving release velocity by 30% and system reliability.
- Delivered dynamic Tableau and Power BI dashboards providing real-time ICU and chronic disease management insights for 5,000+ patients—reducing clinician decision time by 25% and driving a 12% improvement in care delivery KPIs.
- Created and deployed an unsupervised DBSCAN fraud detection model flagging \$4.3M+ in suspicious claims annually, increasing payer audit accuracy by 35%, reducing fraudulent payouts by 20%, and recovering an additional \$1.7M in payments.
- Led Privacy Impact Assessments and embedded HIPAA/HITECH compliance and HITRUST-aligned data governance frameworks across analytics pipelines, maintaining zero data breaches or audit findings over 3 years and securing 50,000+ patient records.
- Mentored 4 junior analysts and championed Agile practices within the team, increasing sprint velocity by 20% and improving cross-functional collaboration.
- Partnered with epidemiologists and population health teams to develop COVID-19 vaccine uptake predictive models leveraging multi-dimensional SDOH data, increasing targeted vaccination effectiveness by 28% across 3 states and supporting public health initiatives.

Healthcare Data Analyst Consultant | KPIT | India

Jan 2021 – Aug 2023

- Automated real-time billing alerts using SQL and Python predictive analytics, reducing claim denial rates by 29% and improving revenue cycle efficiency for a regional health insurer.
- Led deep-dive analyses of chronic disease management programs using ICD-10-coded EMR data and HEDIS metrics, optimizing care pathways and closing quality gaps that improved patient outcomes and compliance.
- Developed a Python-based CMS risk adjustment audit tool, enhancing Medicare Advantage and ACA coding accuracy and driving a 22% increase in compliant reimbursements.
- Spearheaded integration of Salesforce Health Cloud with Epic EMR via HL7 APIs and OAuth2, creating a unified patient view that increased patient outreach engagement by 35%.
- Delivered HIPAA-compliant data science training and security best practices to analytics teams, raising data governance maturity and reducing compliance incidents by 40%.
- Applied advanced NLP techniques to clinical notes for population health risk stratification, enhancing targeted outreach and care management programs.

EDUCATION

M.S. Business Analytics | Sacred Heart University | CT

Dec 2024

Capstone: Advanced predictive modeling of hospital-acquired conditions via MIMIC-III clinical NLP

CERTIFICATIONS

Tableau Desktop Specialist | Google Data Analytics Professional Certificate | IBM Machine Learning for Data Science | Microsoft Excel for Data Analysis